Supplementary Material for Rebuttal (ICML Submission #10645)

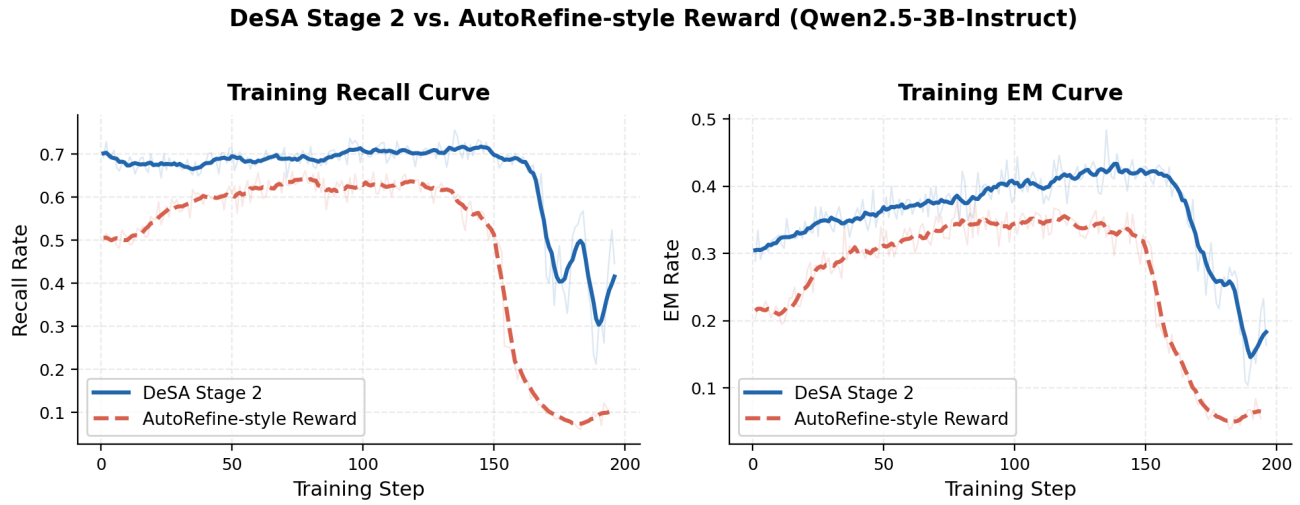

Figure 1: Training recall (left) and EM (right) curves on Qwen2.5-3B-Instruct. DeSA Stage 2 (blue) maintains stable recall ($\sim$0.70) and EM ($\sim$0.42) throughout training. The AutoRefine-style reward (red, dashed) achieves lower peak performance and collapses around step 150, confirming that answer-gated reward shaping leads to premature training collapse.

Table 1: Comparison of DeSA with related methods across four dimensions. **Search Reward Signal**: the intermediate reward supervising search actions; AutoRefine's retrieval reward checks whether the answer appears in the *refined* knowledge output (not raw retrieved documents), activated only when the final answer is wrong; DeSA's search recall reward checks whether retrieved documents contain the answer. **Decoupled from Outcome**: whether the search reward is optimized in a separate stage from the outcome (EM) reward; ✗ indicates joint optimization, creating conflicting gradient pressures. **Credit Assignment**: whether the training signal correctly attributes reward to intermediate search actions; Failed = deficient behaviors unreduced or reinforced; Partially failed = structural flaws remain; Fully resolved = search actions receive unambiguous isolated gradient signal. **Key Limitation**: the primary reason the method cannot fully suppress the deficient search behaviors identified in our analysis.

Method	**Search Reward Signal**	**Decoupled from Outcome**	**Credit Assignment**	**Key Limitation**
Search-R1	None	—	Failed	All three failure modes persist
AutoRefine	Retrieval reward, answer-gated	✗	Partially failed	Retrieval signal masked on correct trajectories
Reward Mixing	Search recall (joint)	✗	Conflicting	Joint optimization creates conflicting gradient pressures
R1-Searcher++	Internal knowledge signal (outcome-based)	✗	Partially failed	Addresses *when* to search, not *how*; credit assignment unresolved
DeSA (Ours)	**Search recall (Stage 1 only)**	✓	**Fully resolved**	—